

052BCA011 – J – 24 – 3148

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**SECOND SEMESTER B.C.A. (NEP) DEGREE EXAMINATION,
JULY/AUGUST 2024**

DATA STRUCTURES (DSC – 1)

Time : 2 Hours]

[Max. Marks : 60

Instructions : 1) Answer any five questions from Part – A.
2) Answer any four questions from Part – B.
3) Answer any three questions from Part – C.

PART – A

Answer any five questions in this Part. Each carries two marks.

1. Define structure.
2. What is pointer ? How to define pointer variable ?
3. Define searching.
4. What is circular queue ?
5. List different types of linked lists and represent singly linked list.
6. Define complete binary tree. (5×2=10)

PART – B

Answer any four questions from this Part. Each carries five marks.

7. Explain dynamic memory allocation functions with syntax.
8. Write an algorithm for binary search technique.
9. What is sorting ? Explain selection sort technique with example.
10. Define stack. Explain PUSH, POP and display operations of stack.
11. Write advantages of linked list. (4×5=20)

[P.T.O.]



PART – C

Answer any three questions from this Part. Each carries ten marks.

12. a) Explain classification of data structure with diagram.
b) List operations on data structure. (6+4=10)
13. Write a program to sort an array of elements using quick sort technique.
14. a) Define queue. Explain ordinary queue operations with example.
b) Convert the following infix notations to prefix notations:
i) $(a + (b - c) + d)/e$
ii) $a \wedge b + (c * d)/e$ (6+4=10)
15. Write an algorithm for creation and deletion of doubly linked list. (3×10=30)
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II Semester B.C.A. Examination, May/June - 2025

(Scheme: SEP)

COMPUTER APPLICATION

Data Structures

Time : 3 Hours

Max. Marks : 80

Instruction : Answer part - A and Part - B.

PART - A

Answer any ten questions :

[10 × 2 = 20]

- 1) Define data structure
- 2) What is a linear data structure
- 3) Define stack. Mention the operations performed on stack.
- 4) What is the difference between queue and stack?
- 5) Write the condition to check overflow in a queue implementation.
- 6) Mention the Applications of linked list.
- 7) Define head pointer in a linked list.
- 8) Define Binary tree with an example.
- 9) What is AVL Tree?
- 10) What is DFS? Mention the data structure used in DFS.
- 11) List any 2 applications of B-Tree.
- 12) What is the time complexity of bubble sort?



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MP-286

15) a) Write a note on memory representation of tree using Array and linked list. [6]

b) With Algorithm explain Inorder, Post order and preorder traversal. [9]

OR

c) Construct a tree from given Inorder, Preorder/post order [8]

i) Pre-order - F A E K C D H G B

In-order - E A C K F H D B G

ii) Post-Order - 7 6 3 5 4 2 1

In-order - 4 2 5 1 6 3 7

d) Write an Algorithm for inserting a node in Binary Search Tree. [7]

i) a) Explain Breadth First Search (BFS) with algorithm and an example. [7]

<https://www.uomonline.com>

b) Mention the difference between BFS and DFS. [4]

c) Explain Adjacency Matrix and Adjacency list [4]

OR

d) Explain the working of selection sort with algorithm. [8]

e) Write a C++ program to perform binary search of an element in an array. [7]



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PART - B

Answer the following questions :

[4 × 15 = 60]

13) a) List and explain the importance of data structures in programming. [7]

b) Explain classification of data structure. [8]

OR

c) Write a note on stack operations [9]



9

d) With algorithm explain the conversion of infix to postfix expression.

[6]

4) a) Write and explain the algorithm for operations in linear queue.

[8]



b) Write a note on Queue applications.

[3]

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c) Explain Circular Queue with its memory representation.

[4]



OR

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d) Explain the insertion of node at beginning and end of singly linked list.

[8]



e) Write a C++ program to delete a node at the end of singly linked list. [7]

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5. What is Graph ? Explain the properties of Graphs in details.
6. Write a 'C' program using function to sort the list of n integer elements into ascending order using bubble sort.
7. What is Stack? Write down the Algorithm for implementing push and pop operation in a stack.
8. What is Tree ? Discuss Binary Tree Traversal Recursive Algorithm in details.
9. What is Sorting? Discuss the types of sorting. Write an algorithm for selection sort for the list of n given numbers.
10. Write short notes on any two of the following :

(a) Application of Data Structure

(b) AVL Tree

(c) Dynamic Memory Allocation

(d) Tree Traversal



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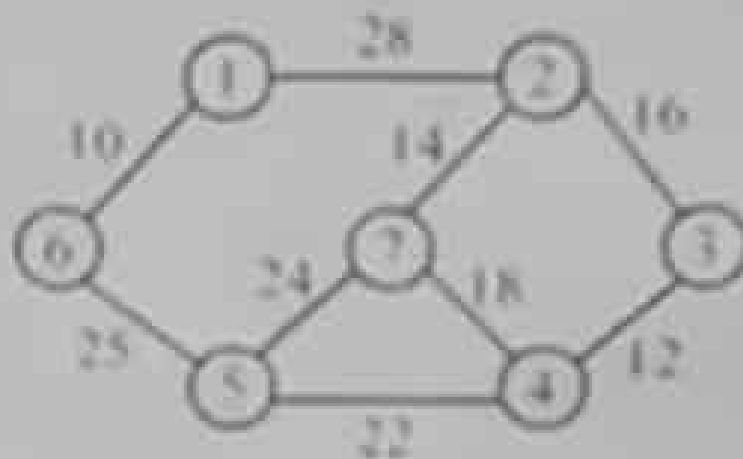


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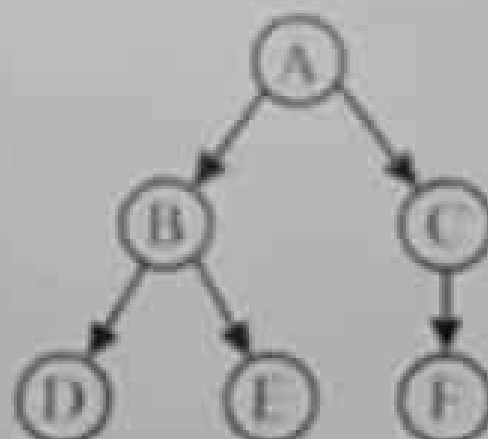


Unit - IV

8. What is minimum cost spanning tree? Compare prim's and kruskal's algorithm with the help of a proper example. Construct the Maximum Spanning Tree (MST) for the given graph using Prim's Algorithm [10]



9. Compare BFS (Breadth First Search) and DFS (Depth First Search) with the help of proper examples. Find BFS and DFS traversal paths for the following directed graph. [10]



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B.C.A. (Semester-II) Examination, 2020

C. PROGRAMMING AND DATA STRUCTURE

| Paper Code : BC-203 |

Time : Three Hours]

[Maximum Marks : 80

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions



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1. What is Data Structure? Discuss the types of data structure in details.



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2. What is Recursive Function? Distinguish between Looping and Recursion with suitable examples.



3. What is Structure. Distinguish between structure and union. Write a program to demonstrate the concept of structure.



4. What is Linked List? Discuss the types of Linked List over Array

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PART – C

Answer any three questions from this Part. Each carries ten marks.

12. a) Explain classification of data structure with diagram
b) List operations on data structure. (6+4=10)
13. Write a program to sort an array of elements using quick sort technique.
14. a) Define queue. Explain ordinary queue operations with example.
b) Convert the following infix notations to prefix notations.
i) $(a + (b - c) * d) / e$
ii) $a * b + (c * d) / e$ (6+4=10)
15. Write an algorithm for creation and deletion of doubly linked list. (3×10=30)
-

6. Define complete binary tree.

(5×2=10)

PART – B

Answer any four questions from this Part. Each carries five marks.

7. Explain dynamic memory allocation functions with syntax.

8. Write an algorithm for binary search technique

9. What is sorting ? Explain selection sort technique with example.

10. Define stack: Explain PUSH, POP and display operations of stack.

11. Write advantages of linked list.

(4×5=20)

[P.T.O.]

Time : 2 Hours]

[Max. Marks : 60

Instructions : 1) Answer any five questions from Part – A.
2) Answer any four questions from Part – B.
3) Answer any three questions from Part – C.

PART – A

Answer any five questions in this Part. Each carries two marks.

1. Define structure.*
2. What is pointer ? How to define pointer variable ?
3. Define searching.*
4. What is circular queue ? *
5. List different types of linked lists and represent singly linked list.*
6. Define complete binary tree.*

(5×2=10)

STUDENT NAME		TOTAL MARKS OBTAINED
CLASS	SUBJECT	
ROLL No.	DATE	

Assignment-1 Unit-1

1. What is Data structure? what are the applications of data structure
2. What are the operations performed in Data structure
3. What is Recursion? what is difference between recursion and iteration
4. Define structure. how to declare and initialize the structure with eg.
5. What is pointer variable? how to declare and initialize pointer variable with eg.
6. what is Union? write the syntax with eg.
7. Explain malloc(), calloc(), realloc() and free() with eg.
8. Explain the memory management functions with eg.
9. what is difference b/w dynamic memory and the static memory allocation.
10. What is difference b/w malloc() and calloc()
11. Explain difference b/w structure and Union
12. what is call by value and Call by reference with eg.

Assignment-2 Unit-2

1. What is stack? Explain the operations on the stack
2. write a program to implement stack operation
3. what are the applications of stack.
4. what is queue what are different types of Queue
5. Explain ordinary queue with all operations
6. Explain double ended queue with all operations
7. Explain circular queue with all operations
8. Explain priority queue with all operations
9. what is bubble sort Explain with eg.
10. what is insertion sort Explain with eg.
11. what is selection sort explain with eg.
12. what is merge sort explain with eg.
13. what is infix, postfix, prefix with eg.
14. Conversion of infix to prefix with eg.
15. Conversion of infix to postfix with eg.